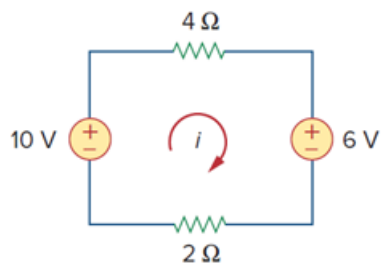


Problem

The loop equation for the circuit in Fig. 3.48 is:

- (a) $-10 + 4i + 6 + 2i = 0$
- (b) $10 + 4i + 6 + 2i = 0$
- (c) $10 + 4i - 6 + 2i = 0$
- (d) $-10 + 4i - 6 + 2i = 0$

Figure 3.48



Step-by-step solution

Step 1 of 5

Refer to figure 10.48 given in the textbook.

Calculate the impedance of the inductor.

$$Z_L = j\omega L$$

Substitute 1 for ω , and 1 for L .

$$\begin{aligned} Z_L &= j(1)(1) \\ &= j1 \, \Omega \end{aligned}$$

Step 2 of 5

Calculate the impedance of the capacitor.

$$Z_C = \frac{1}{j\omega C}$$

Substitute 1 for ω , and 1 for C .

$$\begin{aligned} Z_C &= \frac{1}{j(1)(1)} \\ &= \frac{1}{j} \\ &= -j1 \, \Omega \end{aligned}$$

Step 3 of 5

Consider the voltage source.

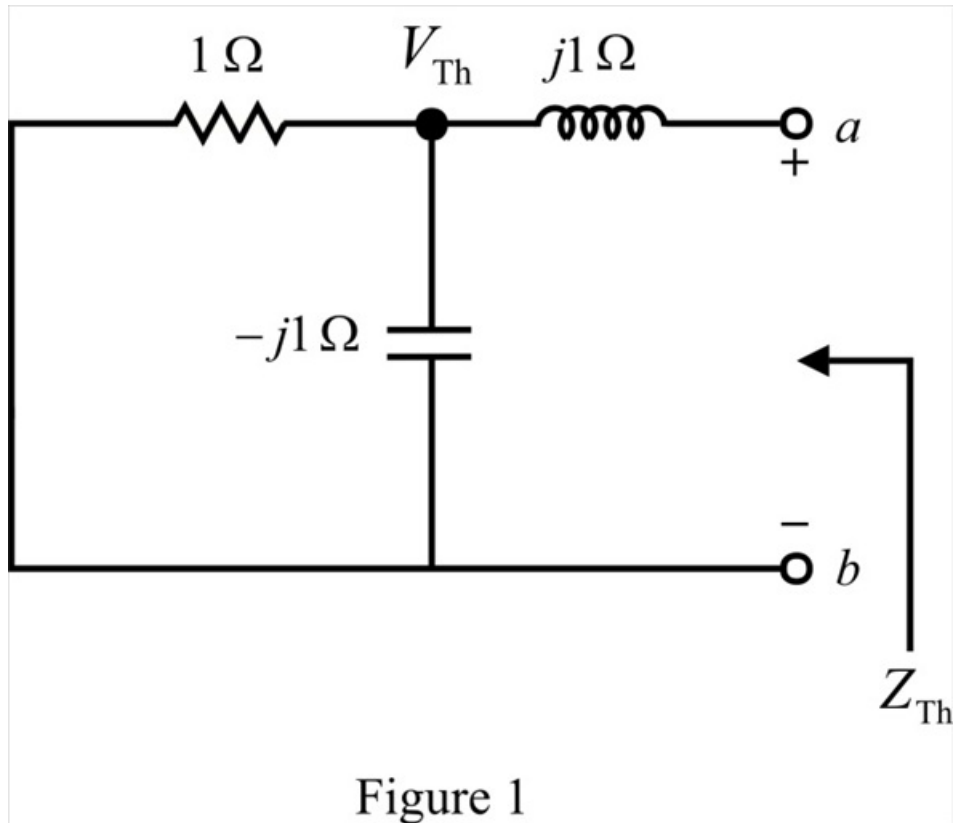
$$v(t) = 5 \cos t \text{ V}$$

Convert $v(t)$ into phasor form.

$$\mathbf{V} = 5 \angle 0^\circ \text{ V}$$

Step 4 of 5

Short circuit the voltage source and redraw the circuit with the impedances of inductor and capacitor as shown in figure 1.



Step 5 of 5

Calculate the Thevenin's impedance Z_{Th} shown in figure 1.

$$\begin{aligned} Z_{Th} &= (1 \Omega \parallel -j1 \Omega) + (j1 \Omega) \\ &= \frac{1 \times (-j1)}{1 - j1} + j1 \\ &= (0.5 + j0.5) \Omega \end{aligned}$$

Therefore, the Thevenin impedance at terminals $a-b$ is $(0.5 + j0.5) \Omega$.

Hence, the correct option is **(c)**.